

Behavioral Risk Factor Surveillance System (BRFSS) 2019

Pradeep Reddy Lopinti¹, Daniel Attah¹, Gayathri Thirumoorthi¹

¹ Department of Computer Science, St. Francis Xavier University; x2024gqa@stfx.ca, x2025ckz@stfx.ca, x2024gpp@stfx.ca

Abstract

Background: This study investigates the prediction of diabetes risk using machine learning models applied to the Behavioral Risk Factor Surveillance System (BRFSS) dataset. The work integrates data preprocessing, class imbalance analysis, feature selection, and explainable artificial intelligence to generate predictive health insights.

Methods: The dataset was subdivided into four groups: sociodemographic, lifestyle, general health, and female-only. Data processing, exploratory analysis, and modeling were performed using the df-analyze framework with Python-based libraries. Seven models were evaluated across five hypotheses: K-Nearest Neighbors, Random Forest, CatBoost, LightGBM, GANDALF, Logistic Regression, and a Dummy baseline classifier. Models were evaluated using accuracy, AUROC, balanced accuracy, F1-score, negative predictive value, positive predictive value, sensitivity, and specificity.

Results: CatBoost demonstrated the strongest overall performance, achieving approximately 87% accuracy and AUROC of 0.85 on the primary BRFSS 2019 dataset. Tree-based models consistently outperformed simpler models across feature groups. In the secondary BRFSS 2020 dataset, CatBoost achieved the highest accuracy in the female-only subgroup, reaching approximately 0.909 accuracy.

Conclusion: The results highlight the usefulness of machine learning methods, particularly CatBoost and LightGBM, for predicting diabetes risk using large population health surveys. Explainable AI analysis also identified key predictors including high blood pressure, high cholesterol, body mass index, and physical inactivity, which align with known diabetes risk factors.

Keywords: Machine learning; feature selection; automated machine learning; df-analyze; health risk behaviors; World Health Organization.

1. Introduction

Diabetes mellitus (DM) is a chronic metabolic disorder characterized by persistent hyperglycemia due to inadequate insulin secretion, impaired insulin action, or both [1-4].

Insulin is a crucial hormone responsible for transporting glucose from the bloodstream into cells for energy metabolism. When insulin function is disrupted, excess glucose accumulates in the blood, leading to a range of symptoms such as excessive thirst (polydipsia), frequent urination (polyuria), fatigue, unintended weight loss, and blurred vision [3,5]. If left untreated or poorly managed, DM can result in severe complications, including cardiovascular disease, kidney failure, neuropathy, and vision loss, significantly reducing quality of life and life expectancy [3,6].

It is one of the most significant global public health challenges and is associated with severe long term complications. According to the World Health Organization, the global prevalence of diabetes has increased substantially over recent decades, placing a growing burden on healthcare systems worldwide [7].

Early identification of individuals at risk of diabetes is essential for implementing preventive interventions and reducing disease progression. Traditional diagnostic approaches rely primarily on clinical tests and physician assessments, which may not always be accessible or cost-effective for

large populations. As a result, there is increasing interest in using large-scale health survey data and computational techniques to support early detection [8].

Recent advances in artificial intelligence and machine learning have enabled the development of automated systems capable of analyzing complex health datasets and identifying hidden patterns associated with disease risk. This study investigates the application of machine learning techniques to predict diabetes using survey-based data from the Behavioral Risk Factor Surveillance System [9]. The primary objective is to evaluate the effectiveness of different learning algorithms and feature selection methods for accurate diabetes classification.

1.1 Overview

Diabetes affects individuals across all age groups and socioeconomic backgrounds and represents a major contributor to global morbidity and mortality. The condition is primarily classified into type 1 diabetes, type 2 diabetes, and gestational diabetes, with type 2 diabetes accounting for the majority of cases. Risk factors include obesity, physical inactivity, an unhealthy diet, smoking, hypertension, and genetic predisposition.

Currently, diabetes diagnosis is based on laboratory tests such as fasting plasma glucose, oral glucose tolerance tests, and glycated hemoglobin (HbA1c) measurements. While these methods are clinically reliable, they require medical infrastructure and regular health monitoring. In many regions, especially under-served communities, access to such diagnostic facilities may be limited.

Large population health surveys such as the Behavioral Risk Factor Surveillance System collect extensive information on demographics, lifestyle behaviors, and health conditions. These datasets provide valuable opportunities to analyze population-level risk factors and develop predictive models. By leveraging such data, it becomes possible to identify individuals who are likely to develop diabetes and support early intervention strategies.

1.2 AI in Healthcare

Artificial intelligence has become an important tool in modern healthcare for disease diagnosis, prognosis, and decision support [10]. Machine learning techniques are capable of learning complex relationships from high-dimensional data and generating accurate predictive models. In recent years, AI has been widely applied to medical imaging, electronic health records, wearable devices, and population health data.

In the context of chronic disease prediction, algorithms such as logistic regression, random forests, support vector machines, k-nearest neighbors, and neural networks have been used to detect patterns associated with disease onset. These methods can analyze interactions among demographic, behavioral, and clinical variables that are difficult to interpret manually. These methods have been widely studied in medical prediction research [9, 11].

For diabetes prediction, AI-based systems have been developed using clinical measurements, genetic data, and lifestyle information. Many studies have reported improved diagnostic accuracy and early detection capabilities compared to traditional risk assessment tools. These developments demonstrate the potential of artificial intelligence to support large-scale disease screening and personalized healthcare.

1.3 Closely related work

Several previous studies have applied machine learning techniques to population based health datasets for diabetes prediction. Research using BRFSS and similar survey datasets has explored the use of random forests, gradient boosting, neural networks, and logistic regression models to identify high-risk individuals [12, 13].

Some studies have reported classification accuracies ranging between 75% and 90% when predicting diabetes status based on demographic and lifestyle variables. Other investigations have focused on feature selection methods to improve model interpretability and reduce redundancy. Commonly

identified risk factors include age, body mass index, physical activity, smoking behavior, and general health status [12].

While these studies demonstrate the feasibility of survey-based diabetes prediction, differences in preprocessing methods, feature selection strategies, and model configurations lead to varying performance outcomes. Therefore, further systematic evaluation using standardized analytical frameworks is necessary. This study builds upon previous work by using the df-analyze framework to conduct comprehensive comparisons of multiple learning algorithms and feature selection approaches.

1.4 Hypotheses

The following hypotheses are investigated in this study:

H1: Full dataset baseline

Models trained using the full BRFSS dataset will produce the highest predictive performance for diabetes classification compared to models trained on smaller feature subsets.

H2 – Demographic predictors

Models trained using demographic variables only will show lower predictive performance compared with the full dataset because they contain less health-related information.

H3 – Lifestyle predictors

Models trained using lifestyle-related variables will provide moderate predictive performance for diabetes prediction but may not perform as well as the full dataset.

H4 – General health predictors

Models trained using general health variables will show stronger predictive performance than demographic or lifestyle subsets because they directly reflect health conditions associated with diabetes.

H5 – Female subgroup

Models trained on the female-only subgroup dataset may produce different predictive performance compared with the full population dataset due to gender-specific health patterns.

2. Materials and Methods

This section describes the dataset, preprocessing procedures, machine learning methods, and evaluation strategies used in this study. All steps are reported in sufficient detail to allow replication and further extension of the work. Publicly available datasets and open-source analytical tools were used throughout the experimental process.

2.1 Dataset Description

The dataset used in this study was derived from the 2019 Behavioral Risk Factor Surveillance System which is a large-scale health survey conducted annually in the United States. The survey collects self-reported information on health behaviors, chronic conditions, and demographic characteristics of adults [9].

The dataset contains approximately 80,000 records after preprocessing and includes variables related to age, sex, body mass index, smoking status, physical activity, alcohol consumption, blood pressure, cholesterol levels, and general health. The target variable, DIABETE4, represents diabetes status and is treated as a binary classification variable, indicating whether an individual has been diagnosed with diabetes.

2.1.1 Data Cleaning and Preprocessing

The following preprocessing steps were applied:

- Removal of incomplete records containing excessive missing values.
- Handling of remaining missing values using deletion or simple imputation.
- Conversion of categorical variables into numerical representations.

- Normalization and scaling of continuous variables where required.
- Detection and removal of duplicate records.
- Verification of valid value ranges for all variables.

After preprocessing, the final dataset was suitable for machine learning analysis.

2.1.2 Target Variable

The target variable DIABETE4 was converted into a binary outcome:

- Class 0: No diabetes
- Class 1: Diagnosed with diabetes

This formulation enables supervised binary classification.

2.2 Machine Learning

The df-analyze framework was used to conduct automated machine learning principles. This tool provides integrated pipelines for data preprocessing, feature selection, model training, hyperparameter optimization, and performance evaluation.

Several classification algorithms were evaluated, including:

- Logistic Regression
- Catboost
- Random Forest
- Gandalf
- Dummy
- LightGBM-based models
- KNN-Nearest Neighbors

Feature selection methods included:

- Wrapper-based selection
- Embedded methods
- Filter-based methods
- Association-based selection

Wrapper-based feature selection evaluates subsets of features by repeatedly training models and selecting combinations that maximize performance. Embedded methods incorporate feature selection during model training, while filter methods rank features based on statistical properties.

Multiple experimental configurations were created to examine the effects of preprocessing strategies and feature selection approaches on classification performance.

2.3 Statistical Analysis

Model performance was evaluated using multiple statistical metrics, including:

- Overall Accuracy
- Balanced Accuracy
- Area Under the ROC Curve (AUC)
- F1-score
- Sensitivity
- Specificity
- Negative Predictive Value
- Positive Predictive Value

These metrics provide complementary perspectives on classification performance, especially in the presence of class imbalance.

Both k-fold cross-validation and standard hold-out validation were applied within the df-analyze frameworks. Cross-validation improves reliability by reducing dependence on a single train-test split. All reported results represent average values across validation folds. Statistical comparisons between models were based on these metrics to identify the most effective learning configurations.

2.4 Secondary Dataset Description

The dataset used in this study was derived from the 2020 Behavioral Risk Factor Surveillance System which is a large-scale health survey conducted annually in the United States. The survey collects self-reported information on health behaviors, chronic conditions, and demographic characteristics of adults [9].

The dataset contains approximately 80,000 records after preprocessing and includes variables related to age, sex, body mass index, smoking status, physical activity, alcohol consumption, blood pressure, cholesterol levels, and general health. The target variable, DIABETE4, represents diabetes status and is treated as a binary classification variable, indicating whether an individual has been diagnosed with diabetes.

This dataset is similar to the primary BRFSS 2019 dataset but represents responses collected in a different survey year, allowing independent validation of the machine learning models.

2.5 Primary Dataset and Secondary Dataset Experimental Configurations

Two experimental configurations were conducted for the primary dataset and secondary dataset. The first configuration evaluated classical and ensemble machine learning models without the inclusion of the deep tabular learner GANDALF. This run included Logistic Regression, Random Forest, LightGBM, CatBoost, K-Nearest Neighbors, and the Dummy baseline classifier.

The second configuration repeated the same experimental setup but included the GANDALF deep learning model. This allowed direct comparison between traditional machine learning models and a deep tabular architecture under the same dataset and preprocessing conditions.

Both configurations used identical preprocessing procedures, feature-selection strategies, and evaluation metrics to ensure fair comparison across models.

3. Results

The df-analyze outputs for five dataset configurations (“runs”) targeting DIABETE4 (binary classification). Results are reported primarily on the holdout set, with supporting training and 5-fold metrics included in each runs.

3.1 Main findings

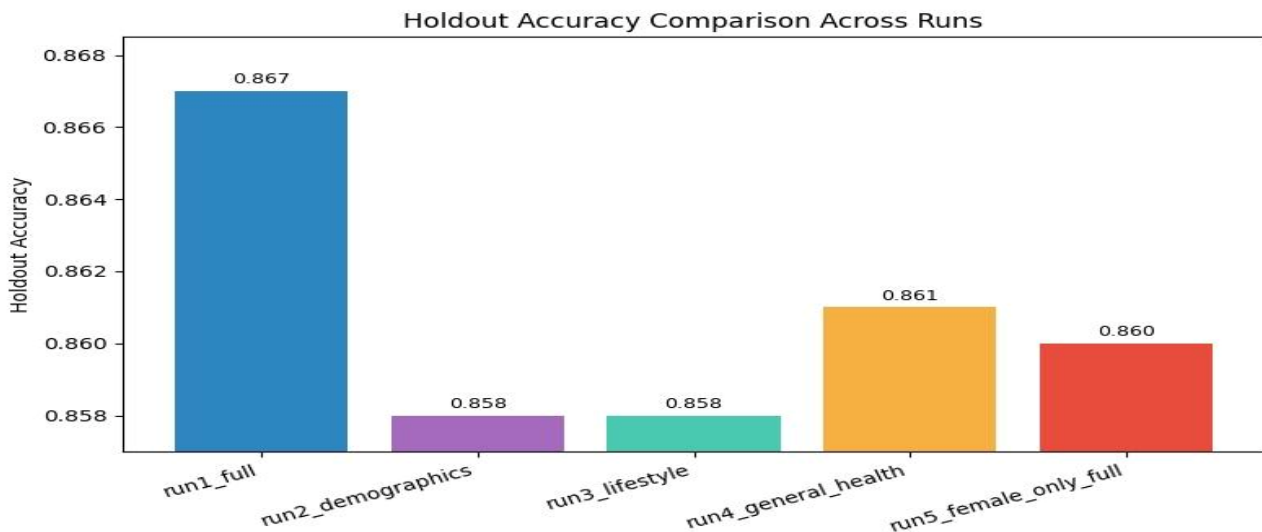
The full-feature run (run1_full) produced the strongest and most consistent predictive performance across models and feature-selection strategies. As shown in Table 1, the best-performing holdout results clustered around Accuracy ≈ 0.866 – 0.867 and AUROC ≈ 0.84 , substantially above the dummy baseline (AUROC = 0.50). However, the best models still showed low sensitivity (~ 0.21 , 0.23) and high specificity (~ 0.98), indicating the classifier was much better at identifying the non diabetes class than the diabetes class (consistent with class imbalance). This behavior is consistent with prior findings in diabetes prediction research [8].

Figure 1. Holdout set performance for the primary run (run1_full).

Abbreviations: *lgbm*, *lightgbm*; *rf*, *Random forest*; *lr*, *logistic regression*; *knn*, *k-nearest neighbors*; *acc*, *accuracy*; *auroc*, *area under the receiver characteristics curve*; *bal-acc*, *balanced accuracy*; *f1*, *f1-score*; *npv*, *negative predictive value*; *ppv*, *positive predictive value*; *sens*, *sensitivity*; *spec*, *specificity*.

Holdout set performance

model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
lgbm	assoc	none	0.867	0.840	0.593	0.618	0.882	0.592	0.210	0.976
lgbm	embed_linear	linear	0.867	0.841	0.599	0.626	0.884	0.579	0.226	0.973
lgbm	none	none	0.866	0.842	0.594	0.619	0.882	0.579	0.214	0.974
rf	assoc	none	0.866	0.832	0.592	0.617	0.882	0.581	0.210	0.975
rf	embed_linear	linear	0.866	0.830	0.604	0.630	0.885	0.566	0.237	0.970
lgbm	wrap	none	0.866	0.838	0.600	0.626	0.884	0.569	0.228	0.971
rf	none	none	0.866	0.831	0.603	0.630	0.885	0.563	0.237	0.969
lr	assoc	none	0.865	0.834	0.591	0.615	0.881	0.568	0.208	0.974
rf	wrap	none	0.865	0.828	0.602	0.627	0.884	0.559	0.234	0.970
lr	none	none	0.865	0.834	0.591	0.615	0.881	0.566	0.208	0.974
lr	embed_linear	linear	0.865	0.835	0.593	0.617	0.882	0.564	0.213	0.973
gandalf	none	none	0.863	0.798	0.563	0.577	0.874	0.563	0.144	0.982
lr	wrap	none	0.862	0.815	0.569	0.585	0.876	0.551	0.159	0.979
gandalf	assoc	none	0.862	0.830	0.549	0.556	0.870	0.567	0.112	0.986
gandalf	embed_linear	linear	0.860	0.808	0.589	0.611	0.881	0.516	0.211	0.967
lgbm	pred	none	0.860	0.820	0.560	0.573	0.873	0.523	0.141	0.979
rf	pred	none	0.859	0.812	0.549	0.556	0.870	0.515	0.116	0.982
lr	pred	none	0.859	0.810	0.563	0.576	0.874	0.511	0.149	0.976
dummy	none	none	0.858	0.500	0.500	0.462	0.858	nan	0.000	1.000
dummy	embed_linear	linear	0.858	0.500	0.500	0.462	0.858	nan	0.000	1.000
dummy	wrap	none	0.858	0.500	0.500	0.462	0.858	nan	0.000	1.000
dummy	pred	none	0.858	0.500	0.500	0.462	0.858	nan	0.000	1.000
dummy	assoc	none	0.858	0.500	0.500	0.462	0.858	nan	0.000	1.000
gandalf	wrap	none	0.857	0.828	0.634	0.655	0.894	0.496	0.323	0.946
gandalf	pred	none	0.852	0.790	0.579	0.597	0.879	0.449	0.200	0.959



3.2 Secondary findings from additional runs of df-analyze

To test whether narrower feature subsets retain predictive signal, four additional runs were evaluated (demographics, lifestyle, general health, and female-only full). The key pattern was that run2_demographics and run3_lifestyle largely collapsed toward the dummy baseline, while run4_general_health and run5_female_only_full showed minor improvements over dummy but still low sensitivity.

3.2.1 Redundancy Aware Feature Selection Results – Secondary run(s)

Several feature-selection strategies were evaluated automatically within the df-analyze framework, including association-based, prediction-based, and embedded selection methods. Across the additional dataset runs, the choice of feature selection method resulted in only small differences in model performance. The overall ranking of models remained similar across configurations, indicating that the predictive signal was largely preserved across different feature subsets. Therefore, feature grouping had a larger effect on performance than the specific selection strategy applied.

3.3 Findings from secondary dataset

To evaluate whether the predictive relationships observed in the primary dataset generalize across survey years, additional experiments were conducted using the BRFSS 2020 dataset. The same df-analyze pipeline, preprocessing procedures, feature-selection strategies, and evaluation metrics were applied to ensure fair comparison between datasets.

The secondary dataset experiments were conducted using four feature-group configurations:

- Demographics
- Lifestyle
- General health
- Female-only subgroup

Across these configurations, tree-based algorithms again demonstrated the strongest performance.

Random Forest achieved the best performance in the demographics group, with accuracy of approximately 0.866. Random Forest also performed best in the lifestyle group, achieving accuracy of approximately 0.867.

In the general health dataset configuration, LightGBM achieved the best performance with accuracy of approximately 0.866.

The female-only subgroup produced the highest predictive performance, where CatBoost achieved accuracy of approximately 0.909, outperforming the other models in that configuration.

These results demonstrate that model performance patterns observed in the 2019 dataset remain largely consistent in the 2020 dataset. Tree-based models continue to outperform simpler algorithms across different feature groupings.

To have a better insight into the prediction model, adaptive error performance was introduced to ascertain the contribution of each features, four extra runs were evaluated (demographics, lifestyle, general health, and female-only). The finding shows that random forest perform best in demographics group with 0.87 accuracy [figure 5], it also perform best in lifestyle group with 0.87 accuracy [figure 6], light-GBM perform best in general health with 0.866 accuracy [figure 7], and catBoost was the best performer in female-only group with 0.91 accuracy [figure 8].

Figure 4. Model performance based on dataset group

Data grouping	Best overall model	Accuracy (acc)
Demographics	Random Forest (rf)	0.866
Lifestyle	Random Forest (rf)	0.867
General health	Light-GBM (lgbm)	0.866
Female only	CatBoost	0.909

Figure 5. Demographics Performance

Holdout set performance

model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
rf	none	none	0.866	0.737	0.501	0.465	0.866	0.625	0.001	1.000
lgbm	embed_linear	linear	0.866	0.742	0.501	0.467	0.866	0.522	0.003	1.000
lgbm	none	none	0.866	0.742	0.501	0.466	0.866	0.538	0.002	1.000
dummy	assoc	none	0.866	0.500	0.500	0.464	0.866	nan	0.000	1.000
rf	embed_linear	linear	0.866	0.734	0.500	0.464	0.866	nan	0.000	1.000
gandalf	pred	none	0.866	0.711	0.500	0.464	0.866	nan	0.000	1.000
gandalf	none	none	0.866	0.721	0.500	0.464	0.866	nan	0.000	1.000
rf	assoc	none	0.866	0.738	0.500	0.464	0.866	nan	0.000	1.000
gandalf	assoc	none	0.866	0.713	0.500	0.464	0.866	nan	0.000	1.000
dummy	embed_linear	linear	0.866	0.500	0.500	0.464	0.866	nan	0.000	1.000
dummy	none	none	0.866	0.500	0.500	0.464	0.866	nan	0.000	1.000
dummy	pred	none	0.866	0.500	0.500	0.464	0.866	nan	0.000	1.000
rf	pred	none	0.866	0.735	0.500	0.464	0.866	nan	0.000	1.000
lgbm	assoc	none	0.866	0.741	0.500	0.465	0.866	0.400	0.000	1.000
catboost	none	none	0.866	0.741	0.500	0.464	0.866	0.000	0.000	1.000
catboost	embed_linear	linear	0.866	0.743	0.502	0.468	0.867	0.459	0.004	0.999
catboost	pred	none	0.866	0.743	0.501	0.466	0.866	0.391	0.002	0.999
catboost	assoc	none	0.866	0.742	0.500	0.464	0.866	0.143	0.000	1.000
lgbm	pred	none	0.866	0.742	0.502	0.468	0.867	0.432	0.004	0.999
gandalf	embed_linear	linear	0.865	0.721	0.501	0.466	0.866	0.231	0.002	0.999

Abbreviations: lgbm, lightgbm; rf, Random forest; lr, logistic regression; knn, k-nearest neighbors; acc, accuracy; auroc, area under the receiver characteristics curve; bal-acc, balanced accuracy; f1, f1-score; npv, negative predictive value; ppv, positive predictive value; sens, sensitivity; spec, specificity.

Figure 6. Lifestyle Performance

Holdout set performance

model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
rf	assoc	none	0.867	0.715	0.511	0.488	0.869	0.543	0.025	0.997
rf	embed_linear	linear	0.866	0.717	0.510	0.486	0.868	0.530	0.023	0.997
rf	none	none	0.866	0.721	0.511	0.489	0.869	0.526	0.026	0.996
lgbm	assoc	none	0.866	0.731	0.509	0.484	0.868	0.526	0.021	0.997
lgbm	pred	none	0.866	0.732	0.509	0.484	0.868	0.515	0.021	0.997
catboost	none	none	0.866	0.730	0.507	0.481	0.868	0.514	0.018	0.997
catboost	assoc	none	0.866	0.723	0.507	0.481	0.868	0.510	0.018	0.997
catboost	embed_linear	linear	0.866	0.729	0.507	0.480	0.868	0.504	0.017	0.997
rf	pred	none	0.866	0.715	0.503	0.470	0.867	0.509	0.006	0.999
dummy	pred	none	0.866	0.500	0.500	0.464	0.866	nan	0.000	1.000
dummy	assoc	none	0.866	0.500	0.500	0.464	0.866	nan	0.000	1.000
dummy	embed_linear	linear	0.866	0.500	0.500	0.464	0.866	nan	0.000	1.000
lgbm	embed_linear	linear	0.866	0.732	0.505	0.475	0.867	0.500	0.011	0.998
dummy	none	none	0.866	0.500	0.500	0.464	0.866	nan	0.000	1.000
gandalf	assoc	none	0.866	0.719	0.500	0.464	0.866	nan	0.000	1.000
gandalf	pred	none	0.866	0.704	0.500	0.464	0.866	0.000	0.000	1.000
catboost	pred	none	0.866	0.732	0.508	0.481	0.868	0.487	0.018	0.997
lgbm	none	none	0.866	0.731	0.509	0.484	0.868	0.481	0.021	0.996
gandalf	embed_linear	linear	0.866	0.708	0.502	0.470	0.867	0.378	0.007	0.998
gandalf	none	none	0.865	0.687	0.500	0.466	0.866	0.216	0.002	0.999

Abbreviations: lgbm, lightgbm; rf, Random forest; lr, logistic regression; knn, k-nearest neighbors; acc, accuracy; auroc, area under the receiver characteristics curve; bal-acc, balanced accuracy; f1, f1-score; npv, negative predictive value; ppv, positive predictive value; sens, sensitivity; spec, specificity.

Figure 7. General Health Performance

Holdout set performance

model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
lgbm	embed_linear	linear	0.866	0.779	0.500	0.464	0.866	1.000	0.000	1.000
lgbm	none	none	0.866	0.781	0.500	0.464	0.866	1.000	0.000	1.000
catboost	assoc	none	0.866	0.780	0.500	0.464	0.866	nan	0.000	1.000
catboost	none	none	0.866	0.781	0.500	0.464	0.866	0.500	0.000	1.000
rf	assoc	none	0.866	0.708	0.500	0.464	0.866	nan	0.000	1.000
rf	embed_linear	linear	0.866	0.770	0.500	0.464	0.866	nan	0.000	1.000
dummy	assoc	none	0.866	0.500	0.500	0.464	0.866	nan	0.000	1.000
dummy	embed_linear	linear	0.866	0.500	0.500	0.464	0.866	nan	0.000	1.000
gandalf	embed_linear	linear	0.866	0.771	0.500	0.464	0.866	nan	0.000	1.000
gandalf	pred	none	0.866	0.777	0.500	0.464	0.866	nan	0.000	1.000
dummy	none	none	0.866	0.500	0.500	0.464	0.866	nan	0.000	1.000
dummy	pred	none	0.866	0.500	0.500	0.464	0.866	nan	0.000	1.000
rf	none	none	0.866	0.769	0.500	0.464	0.866	nan	0.000	1.000
rf	pred	none	0.866	0.766	0.500	0.464	0.866	nan	0.000	1.000
lgbm	assoc	none	0.866	0.778	0.500	0.464	0.866	nan	0.000	1.000
lgbm	pred	none	0.866	0.776	0.500	0.464	0.866	nan	0.000	1.000
catboost	pred	none	0.866	0.780	0.502	0.469	0.867	0.457	0.005	0.999
gandalf	none	none	0.866	0.778	0.500	0.465	0.866	0.273	0.001	1.000
catboost	embed_linear	linear	0.866	0.780	0.501	0.468	0.866	0.400	0.004	0.999
gandalf	assoc	none	0.864	0.764	0.510	0.488	0.869	0.402	0.026	0.994

Abbreviations: lgbm, lightgbm; rf, Random forest; lr, logistic regression; knn, k-nearest neighbors; acc, accuracy; auroc, area under the receiver characteristics curve; bal-acc, balanced accuracy; f1, f1-score; npv, negative predictive value; ppv, positive predictive value; sens, sensitivity; spec, specificity.

Figure 8. Female only Performance

Holdout set performance

model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
catboost	assoc	none	0.909	0.903	0.673	0.726	0.912	0.861	0.354	0.991
catboost	pred	none	0.909	0.905	0.677	0.729	0.913	0.841	0.364	0.990
catboost	none	none	0.909	0.897	0.659	0.713	0.908	0.918	0.322	0.996
lgbm	assoc	none	0.909	0.901	0.673	0.725	0.912	0.851	0.355	0.991
lgbm	none	none	0.908	0.900	0.681	0.732	0.914	0.821	0.374	0.988
lgbm	embed_linear	linear	0.908	0.905	0.676	0.727	0.913	0.839	0.362	0.990
catboost	embed_linear	linear	0.908	0.901	0.666	0.719	0.910	0.875	0.339	0.993
lgbm	pred	none	0.908	0.903	0.670	0.722	0.911	0.856	0.348	0.991
rf	embed_linear	linear	0.907	0.879	0.673	0.723	0.912	0.821	0.357	0.988
rf	assoc	none	0.907	0.879	0.679	0.728	0.914	0.800	0.372	0.986
rf	none	none	0.906	0.880	0.683	0.731	0.915	0.781	0.383	0.984
rf	pred	none	0.906	0.876	0.680	0.728	0.914	0.787	0.375	0.985
gandalf	none	none	0.903	0.869	0.665	0.712	0.910	0.785	0.344	0.986
gandalf	embed_linear	linear	0.902	0.855	0.632	0.680	0.901	0.922	0.267	0.997
gandalf	pred	none	0.901	0.867	0.660	0.706	0.909	0.767	0.335	0.985
gandalf	assoc	none	0.886	0.823	0.673	0.702	0.913	0.595	0.384	0.961
dummy	assoc	none	0.871	0.500	0.500	0.465	0.871	nan	0.000	1.000
dummy	embed_linear	linear	0.871	0.500	0.500	0.465	0.871	nan	0.000	1.000
dummy	none	none	0.871	0.500	0.500	0.465	0.871	nan	0.000	1.000
dummy	pred	none	0.871	0.500	0.500	0.465	0.871	nan	0.000	1.000

Abbreviations: lgbm, lightgbm; rf, Random forest; lr, logistic regression; knn, k-nearest neighbors; acc, accuracy; auroc, area under the receiver characteristics curve; bal-acc, balanced accuracy; f1, f1-score; npv, negative predictive value; ppv, positive predictive value; sens, sensitivity; spec, specificity.

3.4 Adaptive Error Estimation Results

Adaptive error estimation was evaluated using the updated df-analyze framework. This method estimates the expected prediction error for each sample based on the predictive confidence of the model.

Figures 9 and 10 show the relationship between predictive confidence and the expected error rate for the evaluated models. In general, the expected error decreases as the prediction confidence increases. This indicates that predictions made with higher confidence are more reliable.

Figure9. Adaptive error estimation for the BRFSS 2019 dataset

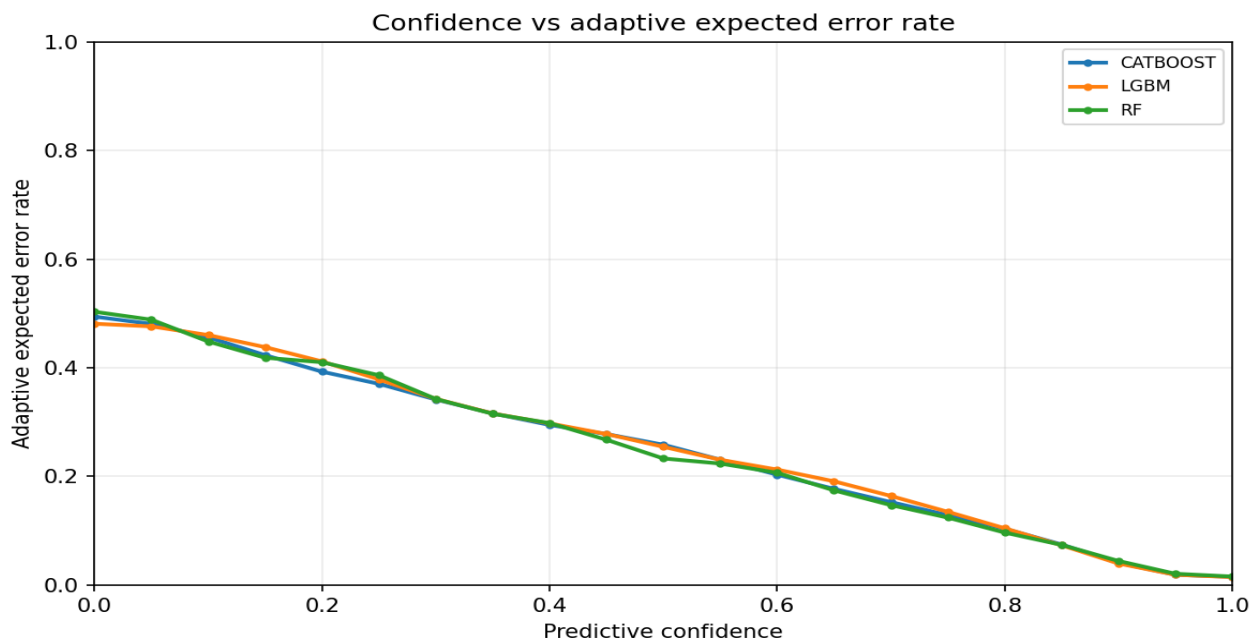
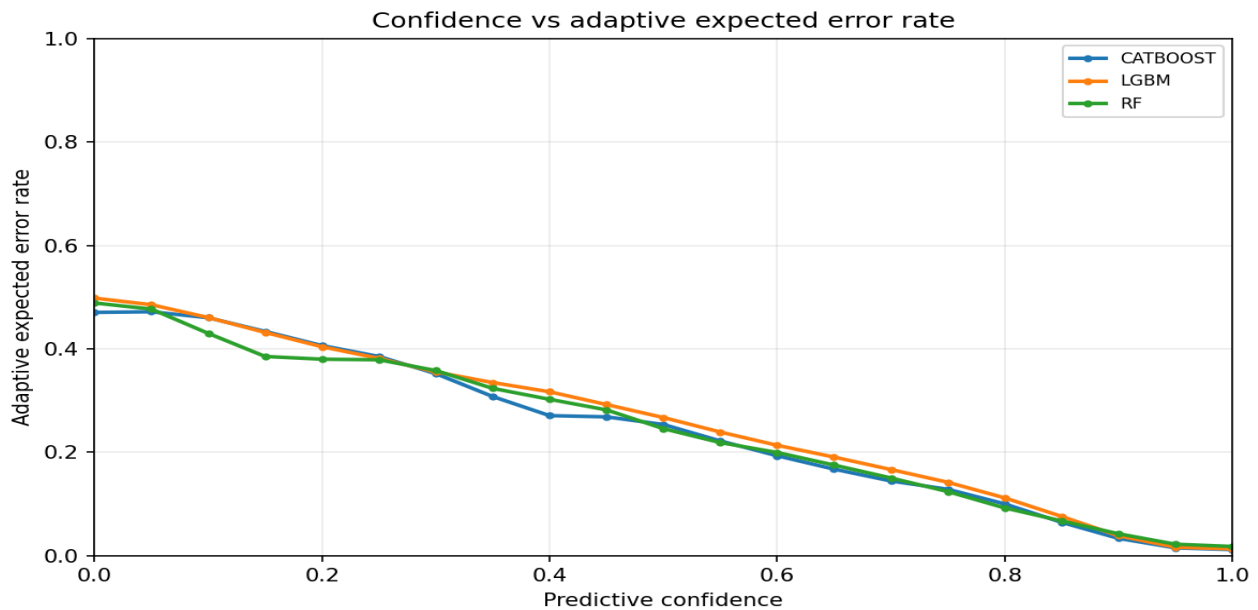


Figure 10. Adaptive error estimation for the BRFSS 2020 dataset



3.5 GANDALF Assessment

From all the models evaluated based on the data group, Gandalf was the least performer except on female-only group with 90% accuracy [figure 8].

Overall, GANDALF showed lower performance compared with the other machine learning models used in this study. Although its overall accuracy was relatively high, this was mainly due to the class imbalance in the dataset. The model showed very low sensitivity, meaning that it rarely identified individuals who had diabetes.

Across the different experimental configurations, GANDALF generally performed worse than LightGBM, Random Forest, and CatBoost. This suggests that the other models were more effective for predicting diabetes using this survey-based dataset.

However, in the female-only dataset configuration, GANDALF achieved performance closer to the best models, reaching approximately 90% accuracy, although its sensitivity was still limited.

Overall, the results show that LightGBM, Random Forest, and CatBoost produced more reliable predictions, while GANDALF did not consistently perform as well in this study.

3.6 Multi-Target Prediction Results

A multi-target prediction run was performed using the BRFSS 2020 dataset. In this experiment, the model was trained to predict both diabetes and prediabetes at the same time, instead of predicting only diabetes as in the earlier runs.

This experiment was carried out using the df-analyze framework with the multi-target configuration on the secondary dataset. In multi-target mode, df-analyze evaluates only CatBoost, K-Nearest Neighbors, and the Dummy baseline model because the other algorithms support only single-target prediction.

The results showed that CatBoost produced the best overall performance among the tested models. Several CatBoost configurations achieved the highest accuracy in the multi-target experiment, reaching approximately 0.868 accuracy. K-Nearest Neighbors and the Dummy baseline produced similar accuracy values but showed weaker AUROC performance. These models were able to achieve high overall accuracy when predicting diabetes and prediabetes together.

However, similar to the earlier experiments, the results still showed higher specificity than sensitivity. This means that the models were better at identifying people without diabetes than correctly identifying people with diabetes or prediabetes. This behavior is mainly due to the class imbalance in the dataset, where non-diabetic cases are much more common.

Overall, the multi-target experiment shows that the models can be used to predict both diabetes and prediabetes using survey data. This type of analysis can help identify people who may be at risk earlier and could be useful for public health monitoring.

Table 3: Holdout set performance for multi-target prediction (BRFSS 2020)

Model	Selection	Embed Select or	Acc	AUROC	Bal-Acc	F1	Hamming-Loss	Multi-Logloss	NPV	PPV	Sens	Spec	Subset-Acc
catboost	none	none	0.868	0.747	0.518	0.498	0.132	0.332	0.716	0.869	0.998	0.037	0.738
catboost	embed_linear	linear	0.868	0.695	0.510	0.485	0.132	0.357	0.662	0.868	0.999	0.021	0.736
catboost	pred	none	0.867	0.668	0.505	0.475	0.133	0.358	0.800	0.867	1.000	0.011	0.734
dummy	assoc	none	0.866	0.500	0.500	0.464	0.134	—	—	0.866	1.000	0.000	0.733
knn	embed_linear	linear	0.866	0.521	0.500	0.464	0.134	—	—	0.866	1.000	0.000	0.733
dummy	embed_linear	linear	0.866	0.500	0.500	0.464	0.134	—	—	0.866	1.000	0.000	0.733
dummy	none	none	0.866	0.500	0.500	0.464	0.134	—	—	0.866	1.000	0.000	0.733
dummy	pred	none	0.866	0.500	0.500	0.464	0.134	—	—	0.866	1.000	0.000	0.733
knn	pred	none	0.866	0.639	0.500	0.464	0.134	—	—	0.866	1.000	0.000	0.733
knn	none	none	0.866	0.521	0.500	0.464	0.134	—	—	0.866	1.000	0.000	0.733
catboost	assoc	none	0.866	0.723	0.502	0.467	0.134	0.341	0.417	0.867	0.999	0.004	0.733
knn	assoc	none	0.864	0.597	0.504	0.476	0.136	—	0.325	0.868	0.993	0.014	0.728

4. Discussion

4.1 Discussion of primary experimental results

The results demonstrate that tree-based models, particularly CatBoost and LightGBM, are the most effective approaches for predicting diabetes risk in the BRFSS dataset, while simpler models fail to adequately detect positive diabetes cases due to class imbalance [Figure 9].

Figure 9. Primary experimental holdout performance

## Holdout set performance										
model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
catboost	assoc	none	0.870	0.848	0.602	0.630	0.884	0.609	0.227	0.976
catboost	none	none	0.869	0.847	0.607	0.635	0.886	0.598	0.240	0.973
catboost	pred	none	0.869	0.847	0.600	0.628	0.884	0.604	0.225	0.976
lgbm	embed_linear	linear	0.869	0.848	0.613	0.641	0.887	0.584	0.255	0.970
catboost	embed_linear	linear	0.868	0.848	0.600	0.627	0.884	0.597	0.226	0.975
lgbm	pred	none	0.868	0.848	0.610	0.639	0.887	0.585	0.250	0.971
lgbm	assoc	none	0.868	0.846	0.607	0.635	0.886	0.585	0.242	0.972
lgbm	none	none	0.868	0.848	0.608	0.635	0.886	0.580	0.244	0.971
rf	none	none	0.866	0.832	0.622	0.649	0.890	0.555	0.281	0.963
rf	assoc	none	0.866	0.836	0.617	0.644	0.889	0.557	0.268	0.965
rf	embed_linear	linear	0.865	0.833	0.624	0.650	0.891	0.543	0.288	0.960
rf	pred	none	0.864	0.832	0.627	0.653	0.892	0.540	0.296	0.958
gandalf	assoc	none	0.860	0.830	0.515	0.495	0.862	0.627	0.034	0.997
gandalf	pred	none	0.859	0.813	0.502	0.467	0.859	0.686	0.005	1.000
gandalf	embed_linear	linear	0.858	0.831	0.502	0.467	0.859	0.610	0.006	0.999
knn	assoc	none	0.858	0.644	0.500	0.462	0.858	nan	0.000	1.000
dummy	assoc	none	0.858	0.500	0.500	0.462	0.858	nan	0.000	1.000
dummy	embed_linear	linear	0.858	0.500	0.500	0.462	0.858	nan	0.000	1.000
dummy	none	none	0.858	0.500	0.500	0.462	0.858	nan	0.000	1.000
dummy	pred	none	0.858	0.500	0.500	0.462	0.858	nan	0.000	1.000
lr	pred	none	0.858	0.546	0.500	0.462	0.858	nan	0.000	1.000
knn	embed_linear	linear	0.858	0.644	0.500	0.462	0.858	nan	0.000	1.000
knn	none	none	0.858	0.644	0.500	0.462	0.858	nan	0.000	1.000
knn	pred	none	0.858	0.644	0.500	0.462	0.858	nan	0.000	1.000
lr	embed_linear	linear	0.858	0.551	0.500	0.462	0.858	nan	0.000	1.000
lr	none	none	0.858	0.546	0.500	0.462	0.858	nan	0.000	1.000
lr	assoc	none	0.858	0.558	0.500	0.462	0.858	nan	0.000	1.000
gandalf	none	none	0.855	0.816	0.498	0.462	0.858	0.035	0.001	0.996

This table presents the holdout set performance of seven machine learning models evaluated under different feature-selection strategies. The models include CatBoost, LightGBM (LGBM), Random Forest (RF), Gandalf, KNN, Logistic Regression (LR), and a Dummy baseline. Performance was assessed using several metrics including accuracy, AUROC, balanced accuracy, F1-score, negative predictive value (NPV), positive predictive value (PPV), sensitivity, and specificity.

4.1.2 Best Performing Models

The results indicate that tree-based models perform the best overall.

CatBoost demonstrates the strongest overall performance among all models with an Accuracy of 0.870, AUROC of 0.848, Balanced Accuracy of 0.60, F1-score of 0.63, NPV of 0.88, PPV of 0.61, Sensitivity of 0.23, and Specificity of 0.97. These results show that CatBoost achieves the highest predictive reliability, particularly in terms of AUROC and F1-score [Figure 9].

LightGBM performs very similarly to CatBoost with an Accuracy of 0.868–0.869, AUROC of 0.848, Balanced Accuracy of 0.61, and F1-score of 0.64. Although slightly lower than CatBoost in some metrics, LightGBM still demonstrates strong predictive performance [Figure 9].

Random Forest also performs competitively with an Accuracy of 0.865–0.866, AUROC of 0.832–0.836, Balanced Accuracy of 0.62, and F1-score of 0.65. Interestingly, Random Forest achieves the

highest sensitivity (~ 0.296) among the top models, indicating better detection of positive diabetes cases [Figure 9].

Both KNN and Logistic Regression perform poorly compared with ensemble methods with Balanced Accuracy of 0.50, Sensitivity of 0.000 in most configurations, and AUROC around 0.54–0.64. These models largely predict the majority class (non-diabetic) and fail to detect diabetic cases.

The Dummy classifier serves as a baseline model with an Accuracy of 0.858, and AUROC of 0.50 [Figure 9]. This confirms that predicting only the majority class produces high accuracy but no predictive discrimination.

4.1.3 Class Imbalance Effect

A clear pattern in the results is the low sensitivity across most models, while specificity remains very high (often above 0.96). This indicates that the dataset is highly imbalanced, with significantly more non-diabetic than diabetic cases. As a result, models easily predict non-diabetes cases (high specificity). They struggle to correctly identify diabetes cases (low sensitivity) [Figure 9].

4.1.4 Effect of Feature Selection

Different feature-selection methods were tested. Association-based selection (assoc), Prediction-based selection (pred), Embedded linear selection (embed_linear), No selection. The results show that feature selection produced only small performance differences. Most top models (CatBoost, LightGBM, Random Forest) maintain similar performance regardless of the feature-selection method.

Ensemble boosting models (CatBoost and LightGBM) provide the best overall performance. Random Forest provides slightly better sensitivity, which may be beneficial for detecting diabetes cases. Traditional models (KNN and Logistic Regression) perform poorly due to dataset complexity and imbalance. Accuracy alone is misleading because the dataset is imbalanced. Balanced accuracy and F1-score provide better evaluation of diabetes prediction models.

4.2 Discussion of Hypotheses

The experiments conducted in this study were designed to evaluate the hypotheses presented in Section 1.4 regarding the predictive ability of different feature groups for diabetes classification.

H1 – Full dataset baseline

The results support the first hypothesis that models trained using the full dataset provide the highest predictive performance. The full dataset configuration achieved the strongest results across the evaluated models. CatBoost reached approximately 0.87 accuracy and AUROC around 0.85, showing that using all available variables improves diabetes prediction.

H2 – Demographic predictors

The demographic-only dataset produced lower predictive performance compared with the full dataset. In several cases, model performance was close to the dummy baseline. This indicates that demographic variables alone provide limited information for identifying diabetes risk.

H3 – Lifestyle predictors

The lifestyle dataset showed moderate predictive performance. Although the models were able to detect some useful patterns, the performance remained lower than the full dataset configuration. This

suggests that lifestyle factors such as smoking behavior, alcohol consumption, and physical activity contribute to diabetes prediction but are not sufficient on their own.

H4 – General health predictors

The general health feature group produced stronger predictive performance than demographic and lifestyle subsets. Variables related to general health condition, blood pressure, and cholesterol were strongly associated with diabetes risk. These variables improved the ability of the models to detect diabetes cases.

H5 – Female subgroup

The female-only dataset produced interesting results. In the secondary BRFSS 2020 dataset, CatBoost achieved the highest accuracy of approximately 0.909 for this subgroup. This suggests that gender-specific health patterns may influence diabetes prediction and that subgroup analysis can sometimes improve model performance.

Overall, the experimental results support the proposed hypotheses. The full dataset provides the strongest predictive performance, while smaller feature groups retain different levels of predictive signal depending on the type of variables included

4.3 Discussion of alternative run(s) of df-analyze

4.3.1 Adaptive Error Performance

The adaptive error plots in Figure 9 and Figure 10 show how prediction confidence is related to the expected error rate.

Across the models, the expected error becomes smaller as prediction confidence increases. This means that predictions with higher confidence are usually more accurate.

Random Forest shows slightly lower error in some confidence ranges. CatBoost performs very similarly to Random Forest and keeps a low error rate as confidence increases. LightGBM follows a similar pattern but sometimes has slightly higher error values.

At low confidence levels, the error rate is high, which means predictions are less reliable. At higher confidence levels, the error becomes much lower. This suggests that the models can make reliable predictions when they are confident.

For diabetes prediction, this is useful because it allows the model to indicate which predictions are more trustworthy.

4.3.2 General Trend

Across all three models, the adaptive expected error rate decreases as predictive confidence increases. This indicates that predictions made with higher confidence tend to be more accurate. When confidence approaches 1.0, the error rate becomes very small (close to 0.02–0.03), showing that highly confident predictions are generally reliable.

4.3.3 Model Comparison

Random Forest (RF) generally shows slightly lower error rates at several confidence levels (particularly between 0.45 and 0.75), indicating slightly better calibration in this range.

CatBoost performs very similarly to Random Forest and maintains consistently low error as confidence increases.

LightGBM (LGBM) follows a similar pattern but occasionally exhibits slightly higher error values compared to the other models.

Overall, the differences between the models are relatively small, suggesting that all three ensemble algorithms demonstrate stable and well-calibrated prediction behavior.

4.3.4 Low Confidence Region

At lower predictive confidence levels (between 0.0 and 0.2), the error rate is relatively high (around 0.40–0.50). This indicates that predictions made with low confidence are more uncertain and prone to misclassification.

4.3.5 High Confidence Region

At higher confidence levels (above 0.8), the adaptive expected error rate drops sharply for all models. This suggests that the models are capable of making highly reliable predictions when they are confident, which is important for healthcare risk prediction systems.

4.3.6 Implication for Diabetes Prediction

In the context of diabetes risk prediction using the BRFSS dataset, the figure suggests that ensemble models such as CatBoost, LightGBM, and Random Forest provide well-calibrated predictions. As predictive confidence increases, the probability of classification error decreases significantly, making these models useful for identifying high-risk individuals with greater reliability. The plot demonstrates that the machine learning models produce more accurate predictions at higher confidence levels, with Random Forest and CatBoost showing slightly better calibration compared to LightGBM.

4.4 Discussion of Feature Selection Results

The relative contribution of each predictor variable to the classification models was assessed using *df-analyze*, a model-agnostic interpretability framework. The framework values quantify each predictor's contribution to the model's output, indicating both the direction and magnitude of its influence on diabetes classification. The analyses were conducted on the best-performing model, and the results were visualized using bar, and box plots. This method provided a consistent and transparent way to interpret complex machine learning algorithms. It also informed potential directions for public health interventions by highlighting the most influential risk and protective factors.

4.5 Redundancy Aware Feature Selection

Redundancy-aware wrapper feature selection was evaluated using the *df-analyze* framework to determine whether a smaller subset of predictors could maintain strong predictive performance while reducing redundant variables.

Across both the primary BRFSS 2019 dataset and the secondary BRFSS 2020 dataset, wrapper-based models produced performance values that were close to the best-performing machine learning models trained using the full feature set. In several configurations, wrapper-based models achieved accuracy values around 0.79–0.82, depending on the learning algorithm used. For example, LightGBM with wrapper selection achieved accuracy values of approximately 0.82, while other models such as stochastic gradient descent achieved slightly lower performance.

Importantly, the wrapper-based models consistently performed better than the dummy baseline classifier, demonstrating that the selected predictor variables contain meaningful information for diabetes risk prediction. Although wrapper-based selection did not always outperform the strongest

tree-based models such as CatBoost, LightGBM, or Random Forest, the performance differences were relatively small.

The wrapper method frequently selected variables related to body mass index, high blood pressure, cholesterol levels, general health status, and physical activity. These predictors are well known risk factors for diabetes and therefore provide reasonable explanations for the model predictions.

Overall, the results suggest that redundancy-aware wrapper feature selection can maintain strong predictive performance while reducing the number of predictor variables, which improves model interpretability and may help identify the most important risk factors associated with diabetes.

Note: The wrapper results used in this section were taken from a separate set of df-analyze results that did not include GANDALF on 2019 and 2020 datasets

4.6 Discussion of GANDALF Results

The Gandalf model shows moderate accuracy but very poor sensitivity. An accuracy of 0.858–0.860, and Sensitivity as low as 0.004–0.006. This suggests that the model rarely identifies positive diabetes cases, even though specificity remains very high [Figure 9].

4.7 Discussion of Secondary Dataset Results

The BRFSS 2020 dataset contains responses from over 400,000 adults collected through telephone surveys across all U.S. states and territories. It includes demographic information, lifestyle behaviors, and health conditions that influence chronic diseases such as diabetes, heart disease, and obesity.

4.8 Literature Comparison

A study on the prediction of diabetes leverages machine learning models, including Logistic Regression, XGBoost, and Random Forest, to predict diabetes using BRFSS data. XGBoost emerged as the top-performing model, achieving an AUC of 0.83. Feature importance was analyzed using SHAP values, highlighting general health, high blood pressure, and BMI as key predictors. This research demonstrates the potential of machine learning models in identifying individuals at risk of diabetes, aiding in early intervention and improved public health outcome [14]

Another study considered Machine Learning (ML) algorithms such as random forest, gradient boosting models, light gradient boosting classifiers, and decision trees, as they are widely used classification algorithms for diabetes prediction. All four ML algorithms were evaluated via performance indicators such as accuracy, F1 score, recall, precision, and AUC-ROC. Comparative analysis of this model suggests that the random forest algorithm outperforms all the remaining classifiers, with the greatest accuracy of 92% on the BRFSS diabetes dataset which is greater than the 3% accuracy reported in existing research [15].

A study used three classification models, Decision Tree, K-Nearest Neighbors (KNN), and Logistic Regression, are implemented and evaluated to determine their predictive performance. The models were trained and tested using a balanced dataset, and their performances are assessed based on accuracy, precision, recall, and F1-score. The results indicate that the Decision Tree, KNN, and Logistic Regression achieve an accuracy of 0.74, 0.72, and 0.75, respectively, with varying strengths in precision and recall. The findings highlight the potential of machine learning in diabetes prediction and suggest future improvements through feature selection and ensemble learning techniques [16].

Finally, an accurate early prediction of diabetes risk may facilitate prevention and timely intervention. A study conducted a comparative analysis of logistic regression (LR), random forest (RF), and Extreme Gradient Boosting (XGBoost) models for diabetes prediction using the Behavioral Risk Factor Surveillance System (BRFSS) 2015 dataset (N = 253,680). The dataset included demographic, lifestyle, and clinical variables. Models were evaluated on accuracy, area under the receiver operating

characteristic curve (ROC AUC), and precision–recall AUC (PR AUC). All models demonstrated strong discriminative performance (ROC AUC = 0.82–0.83), with XGBoost slightly outperforming LR and RF. Feature importance analysis revealed that general health status, high blood pressure, body mass index, cholesterol level, and age were consistent predictors across models. While ML models provided higher predictive performance, logistic regression retained value in interpretability. These findings underscore the potential of integrating machine learning into diabetes risk assessment while maintaining clinical interpretability [17].

4.9 Strengths, Shortcomings, and Future Work

4.9.1 Strengths

This study demonstrates several strengths in the application of machine learning techniques for diabetes prediction using the BRFSS 2019 and 2020 dataset. First, the dataset itself is large and nationally representative, containing responses from hundreds of thousands of individuals. This large sample size improves the reliability and large coverage of predictive models. Additionally, the dataset includes a wide range of demographic, behavioral, and health-related variables such as body mass index (BMI), age, blood pressure, physical activity, and general health status. These features allow machine learning models to capture multiple risk factors associated with diabetes.

Another strength of the study is the use of multiple machine learning algorithms for model comparison. Algorithms such as Random Forest, LightGBM, CatBoost, Logistic Regression, and K-Nearest Neighbors were evaluated using several performance metrics including accuracy, AUROC, specificity, sensitivity, F1-score, and balanced accuracy. Evaluating multiple algorithms provides a comprehensive understanding of model performance and helps identify the most suitable approach for diabetes prediction.

The study also employed validation techniques such as holdout testing and cross-validation to ensure that the models generalize well to unseen data. These evaluation strategies help reduce the risk of overfitting and improve the reliability of the results. Furthermore, feature selection methods were applied to analyze the impact of different subsets of variables on prediction performance, which helps identify the most important risk factors for diabetes.

4.9.2 Shortcomings

Despite these strengths, multiple limitations exist in this study. One major limitation is the class imbalance present in the dataset. In the BRFSS diabetes dataset, the number of non-diabetic cases significantly exceeds the number of diabetic cases. This imbalance often leads to high accuracy but low sensitivity, meaning that the models may fail to correctly identify many positive diabetes cases.

Another limitation is that the BRFSS dataset is based on self-reported survey responses. Self-reported data may contain inaccuracies due to recall bias, misunderstanding of questions, or intentional misreporting by respondents. As a result, some variables may not fully reflect the true health status of participants.

Additionally, the dataset contains only tabular survey data and lacks detailed clinical or laboratory measurements such as blood glucose levels, HbA1c values, or genetic information. The absence of these clinical indicators may limit the predictive power of machine learning models.

Finally, the models used in this study focus primarily on traditional machine learning algorithms. While these methods provide strong baseline performance, more advanced approaches such as deep learning or hybrid models may capture complex relationships in the data more effectively.

4.9.3 Future Work

Future research can address several of the limitations identified in this study. First, techniques for handling class imbalance should be further explored. Methods such as Synthetic Minority Oversampling Technique (SMOTE), or cost-sensitive learning could improve the detection of diabetic cases and increase model sensitivity.

Second, future studies could incorporate additional datasets that include clinical or biomedical information. Combining BRFSS survey data with electronic health records, laboratory measurements, or wearable sensor data may significantly improve prediction accuracy.

Third, more advanced machine learning approaches such as deep neural networks, ensemble learning, and explainable artificial intelligence (XAI) methods could be applied to improve both prediction performance and model interpretability. Explainable AI techniques such as df-analyze feature importance analysis could help identify key risk factors contributing to diabetes.

Finally, future work may also focus on developing real-time or personalized diabetes risk prediction systems that can assist healthcare professionals and public health organizations in early detection and prevention strategies.

5. Conclusions

This study explored the use of machine learning techniques to predict diabetes risk using the Behavioral Risk Factor Surveillance System (BRFSS) dataset. By integrating data preprocessing, class imbalance handling, feature grouping, and explainable artificial intelligence methods, the study demonstrated how machine learning models can generate meaningful insights for public health analysis. The dataset was analyzed across four hypothesis-driven feature groups sociodemographic, lifestyle, general health, and female gender allowing a structured examination of how different health determinants contribute to diabetes risk.

Seven machine learning models were evaluated using a stratified 5-fold cross-validation approach. Among the models tested, CatBoost demonstrated the strongest overall performance, achieving the highest accuracy and AUROC while maintaining competitive performance across other evaluation metrics such as F1-score and balanced accuracy. These results indicate that gradient boosting based ensemble methods are well suited for modeling complex relationships in population health survey datasets. However, the relatively low sensitivity observed across models highlights the ongoing challenge of class imbalance when predicting minority health outcomes such as diabetes.

The analysis also revealed that both clinical indicators and behavioral factors play an important role in diabetes risk prediction. Variables such as high blood pressure, high cholesterol, body mass index, comorbidity burden, and physical inactivity were consistently identified as key predictors. These findings align with existing public health research, confirming that diabetes risk is strongly influenced by a combination of biological, lifestyle, and social determinants of health.

Overall, this study demonstrates the potential of machine learning and automated analytical frameworks to support large-scale health risk prediction using population survey data. In particular, the integration of explainable AI techniques improves transparency and helps identify important risk factors that can guide targeted prevention strategies. Such approaches can support precision public health initiatives by enabling healthcare organizations and policymakers to better identify high-risk populations and design effective intervention programs aimed at reducing the burden of diabetes.

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